

Ecuaciones de la cinemática y la dinámica del manipulador:

$$\begin{aligned}
{}^i{}_{i-1}T &= R_x(\alpha_{i-1})D_x(a_{i-1})R_z(\theta_i)D_z(d_i) \\
R_z(\alpha) &= \begin{pmatrix} c_\alpha & -s_\alpha & 0 \\ s_\alpha & c_\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad ; \quad R_y(\beta) = \begin{pmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{pmatrix} \quad ; \quad R_x(\gamma) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\gamma & -s_\gamma \\ 0 & s_\gamma & c_\gamma \end{pmatrix} \\
c_\alpha &= \cos(\alpha) \quad ; \quad c_\beta = \cos(\beta) \quad ; \quad c_\gamma = \cos(\gamma) \\
s_\alpha &= \sin(\alpha) \quad ; \quad s_\beta = \sin(\beta) \quad ; \quad s_\gamma = \sin(\gamma) \\
{}^{i+1}\omega_{i+1} &= {}^i{}_{i+1}R^i\omega_i + \dot{\theta}_{i+1} {}^{i+1}\hat{Z}_{i+1} \\
{}^{i+1}\dot{\omega}_{i+1} &= {}^i{}_{i+1}R^i\dot{\omega}_i + {}^i{}_{i+1}R^i\omega_i \times \dot{\theta}_{i+1} {}^{i+1}\hat{Z}_{i+1} + \ddot{\theta}_{i+1} {}^{i+1}\hat{Z}_{i+1} \\
{}^{i+1}v_{i+1} &= {}^i{}_{i+1}R[{}^iv_i + {}^i\omega_i \times {}^iP_{i+1}] + \dot{d}_{i+1} {}^{i+1}\hat{Z}_{i+1} \\
{}^{i+1}\dot{v}_{i+1} &= {}^i{}_{i+1}R[{}^i\dot{v}_i + {}^i\dot{\omega}_i \times {}^iP_{i+1} + {}^i\omega_i \times ({}^i\omega_i \times {}^iP_{i+1})] + 2{}^{i+1}\omega_{i+1} \times \dot{d}_{i+1} {}^{i+1}\hat{Z}_{i+1} + \ddot{d}_{i+1} {}^{i+1}\hat{Z}_{i+1} \\
{}^iv_{Ci} &= {}^iv_i + {}^i\omega_i \times {}^iP_{Ci} \\
{}^{i+1}\dot{v}_{Ci+1} &= {}^{i+1}\dot{\omega}_{i+1} \times {}^{i+1}P_{Ci+1} + {}^{i+1}\omega_{i+1} \times ({}^{i+1}\omega_{i+1} \times {}^{i+1}P_{Ci+1}) + {}^{i+1}\dot{v}_{i+1} \\
{}^{i+1}F_{i+1} &= m_{i+1} {}^{i+1}\dot{v}_{Ci+1} \\
{}^{i+1}N_{i+1} &= {}^{Ci+1}I_{i+1} {}^{i+1}\dot{\omega}_{i+1} + {}^{i+1}\omega_{i+1} \times {}^{Ci+1}I_{i+1} {}^{i+1}\omega_{i+1} \\
{}^if_i &= {}^i{}_{i+1}R^{i+1}f_{i+1} + {}^iF_i \\
{}^in_i &= {}^i{}_{i+1}R^{i+1}n_{i+1} + {}^iN_i + {}^iP_{Ci} \times {}^iF_i + {}^iP_{i+1} \times {}^i{}_{i+1}R^{i+1}f_{i+1} \\
\tau_i &= {}^if_i^T {}^i\hat{Z}_i \\
\tau_i &= {}^in_i^T {}^i\hat{Z}_i \\
\tau &= \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\Theta}} \right) - \frac{\partial \mathcal{L}}{\partial \Theta} \quad ; \quad \mathcal{L}(\Theta, \dot{\Theta}) = k(\Theta, \dot{\Theta}) - u(\Theta) \\
k_i &= \frac{1}{2} m_i v_{Ci}^T v_{Ci} + \frac{1}{2} {}^i\omega_i^T {}^{Ci}I_i {}^i\omega_i \quad ; \quad k = \sum_{i=1}^n k_i \quad ; \quad k(\Theta, \dot{\Theta}) = \frac{1}{2} \dot{\Theta}^T M(\Theta) \dot{\Theta} \\
u_i &= -m_i {}^0g^T {}^0P_{Ci} + u_{ref_i} \quad ; \quad u = \sum_{i=1}^n u_i \\
\tau &= M(\Theta) \ddot{\Theta} + V(\Theta, \dot{\Theta}) + G(\Theta) + F(\Theta, \dot{\Theta}) \\
\mathcal{F} &= M_x(\Theta) \ddot{\mathcal{X}} + V_x(\Theta, \dot{\Theta}) + G_x(\Theta) + F_x(\Theta, \dot{\Theta}) \\
\tau &= M(\Theta) \ddot{\Theta} + B(\Theta) [\dot{\Theta} \dot{\Theta}] + C(\Theta) [\dot{\Theta}^2] + G(\Theta) \\
\tau &= J^T(\Theta) M_x(\Theta) \ddot{\mathcal{X}} + B_x(\Theta) [\dot{\Theta} \dot{\Theta}] + C_x(\Theta) [\dot{\Theta}^2] + G(\Theta) \\
\tau &= J^T(\Theta) \mathcal{F} \quad ; \quad v = J(\Theta) \dot{\Theta} \\
{}^AI &= \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{pmatrix} \quad ; \quad I_{xx} = \iiint_V (y^2 + z^2) \rho \, dv \quad ; \quad I_{xy} = \iiint_V xy \rho \, dv \\
A^\# &= (A^T A)^{-1} A^T \quad ; \quad {}^AT = {}^AT {}^BT {}^CT
\end{aligned}$$